

Jonathan Cruz

SENIOR SOFTWARE DEVELOPER · AI-ACCELERATED ENGINEERING

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PROFILE

"Senior developer with 10+ years across the full software lifecycle: design, prototyping, development, testing, deployment, and support. AI agent in my daily workflow for over a year — Claude Code with sub-agents, custom skills, MCP integrations, and project-level specifications applied to real production code. Strong React + TypeScript production UIs with concurrent and event-driven backend systems (Node.js, NestJS, PostgreSQL, Redis).

Track record of mentoring engineers, leading code reviews, and shipping under real production traffic in regulated environments."

PROFESSIONAL EXPERIENCE

SELECTED ROLES & CONTRIBUTIONS

FINTECH PLATFORM

Sydecar

FULL-STACK SOFTWARE ENGINEER

AI-Accelerated Daily Workflow

CLAUDE CODE

Daily user of Claude Code with sub-agent orchestration for parallel exploration and review, custom project skills for repeatable tasks, MCP server integrations, pre/post-tool hooks for guardrails, and project-level specifications. Apply AI to generate unit tests, draft PR summaries, compose migrations, and assist refactors — measurable impact on cycle time and PR throughput.

CLAUDE CODE AI AGENTS MCP SUB-AGENTS

Production UIs

REACT + TYPESCRIPT

Build admin dashboards, internal operational tools, and stakeholder-facing UIs across an Nx monorepo of dozens of React/Next.js apps. Modern React patterns (hooks, context, Redux-style state), TanStack Query, Zod runtime validation, full TypeScript discipline end-to-end.

REACT TYPESCRIPT NEXT.JS ZOD

Concurrency & Multi-threading

RACE-RESISTANT FLOWS

Designed race-resistant verify-identity flow with re-check inside to handle "email-free-at-blur → taken-at-submit" edge case (409 EMAIL_NOW_EXISTS). Standardized webhook deduplication via (unique) with PROCESSING/PROCESSED/REJECTED state machine, and API-level idempotency interceptor with 48h TTL.

CONCURRENCY IDEMPOTENCY RACE CONDITIONS

Event-Driven Systems

BULLMQ + REDIS

Standardized event delivery using BullMQ Pro with exponential backoff (5 retries / 3s base in ledger; 20 retries / 10s base in pubsub). Outbox pattern: persist resolution first, enqueue after — jobs idempotent and survive enqueue failure. Global per-deal ordering via Redis INCR counters with deterministic predecessor lookup.

EVENT-DRIVEN BULLMQ REDIS OUTBOX

SQL & Performance

POSTGRESQL DEEP

Operate Postgres via Prisma with schema design, indexing, transaction isolation, and query performance tuning across dozens of microservices. Auxiliary tables with deterministic ordering, deep transactional consistency for compliance-critical writes.

POSTGRESQL PRISMA SQL TRANSACTIONS

Observability

DATADOG + STRUCTURED LOGGING

Integrated Datadog for tracing, metrics, and alerting on critical paths. Designed audit-friendly state machines and append-only auxiliary tables — reconstructable event chains for production debugging and compliance review.

DATADOG OBSERVABILITY LOGGING TRACING

Security & PII

OAuth + AUDIT

Implemented OAuth 2.0 flows with refresh-token handling and encrypted credential storage. PII-aware data handling across KYC/EDD, UBO verification, FinCEN integration. Production secure coding practices and OWASP-aligned reviews.

OAuth 2.0 PII SECURITY OWASP

Testing

JEST + PLAYWRIGHT + CUCUMBER BDD

Built full E2E test suite (Playwright + Cucumber BDD) for QuickBooks integration with mock services. Unit, integration, and E2E tests integrated into CI on every PR.

JEST PLAYWRIGHT CUCUMBER BDD E2E

MAY 2023 — PRESENT
SAN FRANCISCO, CA (REMOTE)

B2B CLOUD PLATFORM Zax SR SOFTWARE ENGINEER JAN 2019 — MAY 2023 SÃO PAULO	Full-Stack Product REACT + NEXT.JS + GRAPHQL Cloud Deployment AWS + DOCKER CI/CD GITHUB ACTIONS Built from scratch a B2B wholesale ordering platform with React Native (mobile), React/Next.js (web), and a GraphQL API layer. Integrated payment flows and credit-limit validation across the stack. REACT NATIVE NEXT.JS GRAPHQL TYPESCRIPT Deployed and operated full cloud infrastructure on AWS (EC2, ECS, S3, Lambda, VPCs) with serverless React/Next.js apps. Containerized services with Docker for consistent dev-to-prod parity. AWS ECS DOCKER LAMBDA Configured end-to-end pipeline with GitHub Actions, Fastlane, and Code Climate for automated quality gates. Reduced release cycle from weeks to days. GITHUB ACTIONS FASTLANE CI/CD
REAL-TIME PLATFORM Netlolo SR SOFTWARE ENGINEER SEP 2018 — JAN 2019 SÃO PAULO	Real-time SDK CONCURRENT CONNECTIONS Consumer Product REACT NATIVE + WEB3 Zero-downtime CI/CD CONTINUOUS DEPLOYMENT Designed and shipped a chat SDK with React Native and React/Redux on top of XMPP and a Java backend — persistent connections, concurrent connection management, message ordering, low-latency delivery for embedded business use. XMPP REAL-TIME SDK CONCURRENCY Built a mobile and web product bringing cryptocurrency to mainstream users with React Native and TypeScript, integrating Web3 smart contracts on Ethereum. REACT NATIVE TYPESCRIPT WEB3 Built CI/CD automations ensuring continuous deployment without service interruption. PostgreSQL with Hasura as the GraphQL layer. POSTGRESQL HASURA CI/CD
MICROSERVICES PLATFORM Escale SR SOFTWARE ENGINEER JAN 2018 — SEP 2018 SÃO PAULO	Mentorship TECH TALKS LEAD Multi-DB Operations PERSISTENCE SELECTION Led internal Tech Talks on RESTful design, Functional Programming, and Microservices architecture, coaching engineers across product teams. Established service-boundary and API-contract patterns adopted across the engineering org. MENTORSHIP TECH TALKS MICROSERVICES Built the product foundation with React, Redux, and a Node.js/TypeScript backend. Selected MySQL, PostgreSQL, MongoDB per service responsibility with hands-on operational ownership. REACT NODE.JS MONGODB CIRCLECI
BACKEND MODERNIZATION PraValer PL SOFTWARE ENGINEER JUN 2017 — JAN 2018 SÃO PAULO	Legacy Modernization PHP → NODE SQL Performance Tuning INDEXES & SCHEMAS Led migration of a legacy PHP origination platform to Node.js/Express for a student loans product. Modernized API surface while preserving downstream compatibility — same pattern applicable to modernizing legacy .NET Framework codebases. NODE.JS EXPRESS LEGACY MODERNIZATION Restructured PostgreSQL transactional tables with improved schemas and indexes, eliminating N+1 queries and reducing query times by an order of magnitude. MongoDB for unstructured logs. POSTGRESQL MONGODB QUERY OPTIMIZATION

CORE SKILLS

AI-ACCELERATED DEVELOPMENT

Claude Code (daily), Sub-agent orchestration, MCP servers, Custom skills/hooks, AI-generated tests, AI-assisted refactors, Prompt engineering, LLM API integration

FRONTEND

React (hooks, context, Redux), TypeScript, Next.js, React Native, HTML5, CSS3, Web APIs, JSON, REST

BACKEND & APIS

Node.js, NestJS, ts-rest, Zod, GraphQL, RESTful design, OAuth 2.0

DATABASES

PostgreSQL (deep), MySQL, MongoDB, Redis, SQL performance tuning, Indexing, Transactions

CONCURRENCY & DISTRIBUTED

Multi-threading concepts, Race-resistant flows, Idempotency, BullMQ, Outbox pattern, Event-driven workflows

CLOUD & DEVOPS

AWS (EC2/ECS/S3/Lambda/VPC), GCP, Docker, CI/CD (GitHub Actions, Fastlane, CircleCI), Nx Monorepo

OBSERVABILITY & SECURITY

Datadog, Structured logging, Tracing, Audit trails, OAuth 2.0, PII handling, OWASP basics

QUALITY

Jest, Playwright, Cucumber BDD, TDD, Unit + Integration + E2E

LANGUAGES

PORTUGUESE

Native

ENGLISH

C1/C2

EDUCATION

Anhanguera

BACHELOR OF TECHNOLOGY (B.TECH.) IN INFORMATION TECHNOLOGY ·
JAN 2016 — JUL 2018